

Information Retrieval

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Exercise 1.1 [★]

Draw the **inverted index** that would be built for the following document collection. (See Figure 1.3 for an example.)

Doc 1 new home sales top forecasts

Doc 2 home sales rise in july

Doc 3 increase in home sales in july

Doc 4 july new home sales rise

Exercise 1.2 [★]

Consider these documents:

Doc 1 breakthrough drug for schizophrenia

Doc 2 new schizophrenia drug

Doc 3 new approach for treatment of schizophrenia

Doc 4 new hopes for schizophrenia patients

- a. Draw the **term-document incidence matrix** for this document collection.
- b. Draw the **inverted index** representation for this collection, as in Figure 1.3 (page 7).

Exercise 1.3 [★]

For the document collection shown in Exercise 1.2, what are the **returned results** for these queries:

- a. schizophrenia AND drug
- b. for AND NOT(drug OR approach)

Exercise 1.4 [★]

For the queries below, can we still run through the intersection in time $O(x+y)$, where x and y are the lengths of the postings lists for Brutus and Caesar? If not, what can we achieve?

- a. Brutus **AND NOT** Caesar
- b. Brutus **OR NOT** Caesar